

Exp no :08

Date:

Use Steg-Expose to Detect Hidden Data in Images

Description

StegExpose is a specialized tool used for **steganography analysis**. It works by evaluating the **statistical properties** of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

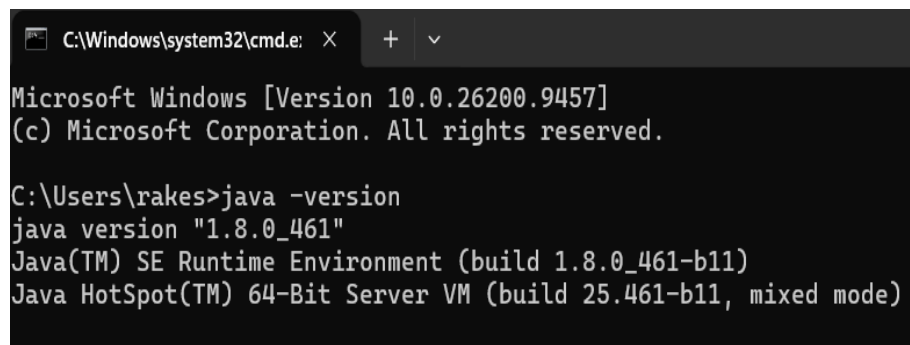
Prerequisites

- Java Runtime Environment (JRE): StegExpose is a Java-based application and requires JRE to run.
- StegExpose Tool: Download the latest .jar file from the official StegExpose GitHub repository.

Step-by-Step Process

1. Download and Set Up StegExpose

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.

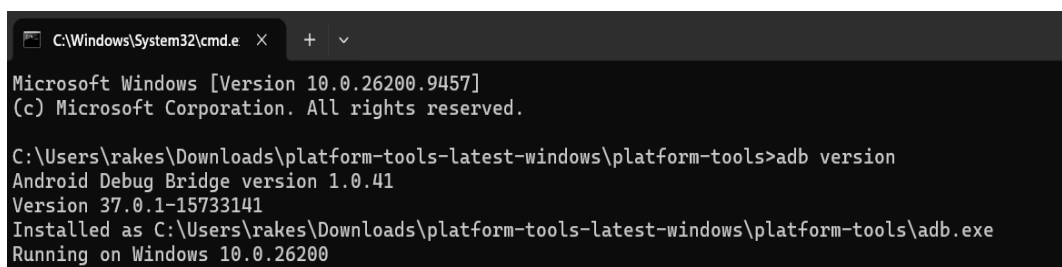


```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rakes>java -version
java version "1.8.0_461"
Java(TM) SE Runtime Environment (build 1.8.0_461-b11)
Java HotSpot(TM) 64-Bit Server VM (build 25.461-b11, mixed mode)
```

2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.

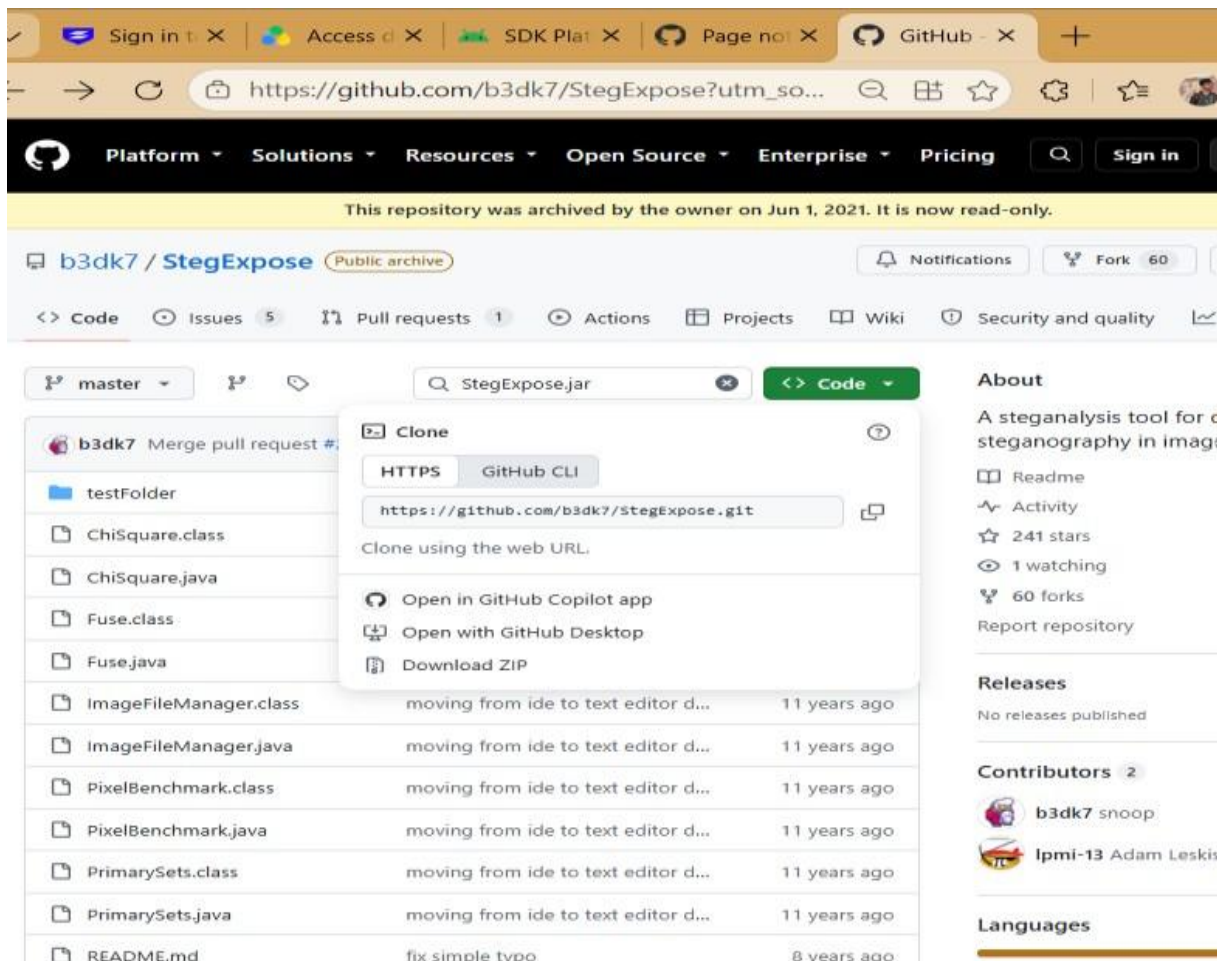
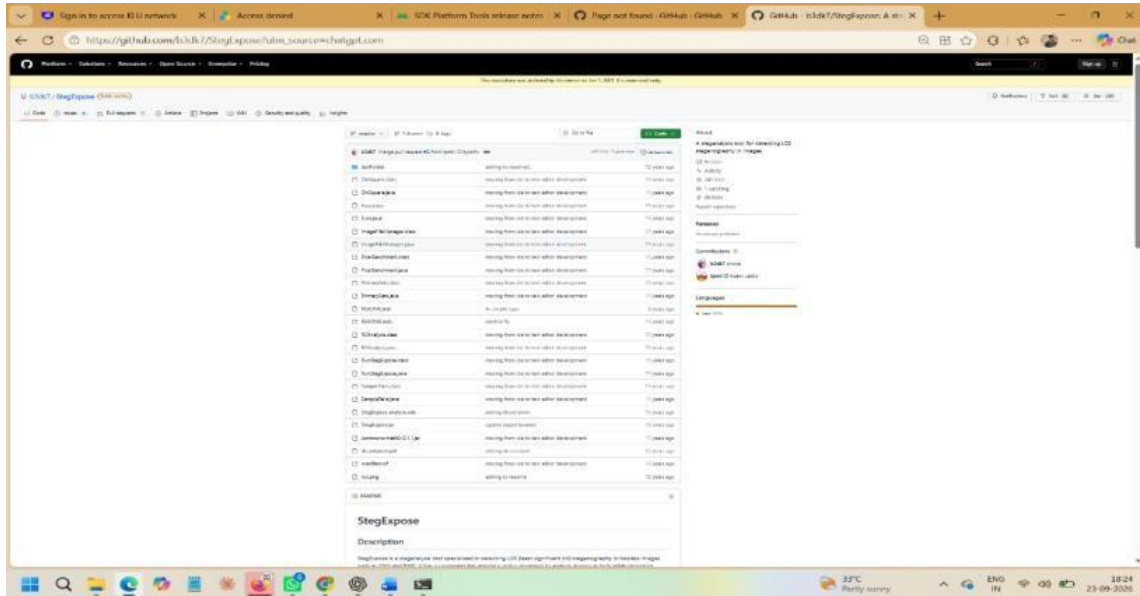


```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rakes\Downloads\platform-tools-latest-windows\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\rakes\Downloads\platform-tools-latest-windows\platform-tools\adb.exe
Running on Windows 10.0.26200
```

3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).



4. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

Command Structure:

```
java -jar StegExpose.jar <image_file_path>
```

Example:

```
java -jar StegExpose.jar test_image.png
```

5. Analyze the Output

- StegExpose calculates a **"suspect" score** ranging from 0 to 1. **The higher the score, the more likely steganography is present.**

Score Range	Interpretation (Suggested Thresholds)
Less than 0.2	Image is considered clean (no hidden data detected).
0.2 - 0.3	Possibly some hidden data is present.
Above 0.3	Likely that steganography is present.

- **Example Output Analysis:**

```
java -jar StegExpose.jar suspect_image.png
```

Analyzing suspect_image.png...

Result: 0.4

Steganography likely present

```
C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder default default steganalysisOfTestFolder

C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>dir steganalysisOfTestFolder*
Volume in drive C is Windows-SSD
Volume Serial Number is DC08-8AC2

Directory of C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master

23-09-2026  18:49                2,255 steganalysisOfTestFolder
             1 File(s)                2,255 bytes
             0 Dir(s) 283,839,115,264 bytes free

C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>
```

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.20200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rakes>cd /d "C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master"

C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>dir
Volume in drive C is Windows-SSD
Volume Serial Number is DC68-BAC2

Directory of C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master

23-09-2026 18:46 <DIR> .
23-09-2026 18:38 <DIR> ..
23-09-2026 18:38 4,690 ChiSquare.class
23-09-2026 18:38 9,188 ChiSquare.java
23-09-2026 18:38 1,599,627 commons-math3-3.1.1.jar
23-09-2026 18:38 682,689 dissertation.pdf
23-09-2026 18:38 744 Fuse.class
23-09-2026 18:38 843 Fuse.java
23-09-2026 18:38 907 ImageFileManager.class
23-09-2026 18:38 1,902 ImageFileManager.java
23-09-2026 18:38 85 manifest.mf
23-09-2026 18:38 377 PixelBenchmark.class
23-09-2026 18:38 1,818 PixelBenchmark.java
23-09-2026 18:38 1,371 PrimarySets.class
23-09-2026 18:38 2,403 PrimarySets.java
23-09-2026 18:38 4,657 README.md
23-09-2026 18:38 4,739 README.md~
23-09-2026 18:38 83,762 rec.png
23-09-2026 18:38 8,959 RSAnalysis.class
23-09-2026 18:38 20,745 RSAnalysis.java
23-09-2026 18:38 5,202 RunStegExpose.class
23-09-2026 18:38 6,822 RunStegExpose.java
23-09-2026 18:38 3,322 SamplePairs.class
23-09-2026 18:38 6,348 SamplePairs.java
23-09-2026 18:38 10,621,897 StegExpose.analysis.ods
23-09-2026 18:38 3,060,550 StegExpose.jar
23-09-2026 18:38 <DIR> testFolder
23-09-2026 18:46 1,286 testFolder.lnk
25 File(s) 16,134,915 bytes free
3 Dir(s) 283,839,598,592 bytes free

C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder
stego_6666458261_e455d262b5_x.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego_6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego_6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.

C:\Users\rakes\Downloads\StegExpose-master\StegExpose-master>
```

6. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:\

Command Structure:

java -jar StegExpose.jar <folder_path>

7. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

Command:

```
File name:Abuse stego threshold:Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,Rs analysis,Fusion (mean)
clean_7221100570_82a10bc040_z.png,false,13416,0.11329240178885477,0.1423569884449125,null,0.0487352556189453034,0.10146154880774011
clean_7222762380_2688184c8e.png,false,7391,NaN,0.7411080131793276,11,0.24006602550881854,0.25861310226721124,0.16622637596448026
clean_7228174652_8f743dc3dd_z.png,false,2469,0.431137468869920087,0.007354582275781215,null,0.01607070559017578,0.0318175855785933
clean_722818722_1c725dff3e_z.png,false,14194,0.116140905612938,0.0021023440017611224,null,0.09988112328129458,0.07279812429966454
clean_7231021538_f5db2a35a_z.png,false,8198,0.009376727810800474,0.018013031823677696,null,0.06105308887385262,0.043834403187130026
clean_7232193200_c2f38e0f25_z.png,false,22569,0.2573883057346056,0.002651049049337065,0.09692103046233898,0.10675110693744194,0.11592787304593088
clean_7232204240_d30b6d6986_z.png,false,25126,NaN,0.0090908057116066,0.18017057666719039,0.21013951225094777,0.133356048232415
clean_7232206610_b9cfded120_z.png,false,10190,0.0284808431325644658,0.03596658763921852,0.080529823771301,0.08440531345457021,0.05734551084768359
clean_723220862_34a2c69109_z.png,false,18800,NaN,0.0125268694583774,0.1321785093498258,0.14740747737848,0.097508164984100398
clean_7234473324_bbaa82b5bd_z.png,false,1403,0.0163592591872635,0.062057618747316554,null,0.03616633877410855,0.01840472618129262
clean_7235558256_3099866753.png,false,21450,0.17546102698982005,0.170494348209659,null,0.20167053078214175,0.18469366419764324
clean_7235972310_8c45238da5.png,false,10882,0.18496458785274301,0.001567093035488728,null,0.00728431990551831,0.095880167725127103
clean_7237798856_86878c8823_z.png,false,3654,0.07330286436492333,0.569594619348558E-4,null,0.06965810797476018,0.04791087393387279
stego_6666458261_e455d262b5_z.png,true,114785,NaN,0.06767821105235003,0.72020221289508,0.738812533683016,0.5116576506419387
stego_6672108499_85c582a7f9.png,true,137047,NaN,0.9999905976238456,null,0.8608770381271956,0.9349338138755205
stego_6672542201_532f70bffe.png,true,67141,NaN,0.09371291967741853,0.7812415352997489,0.7680715936114042,0.5476753495298875
```

java -jar StegExpose.jar --help

8. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.